GOVERNMENT OF INDIA • MINISTRY OF SCIENCE & TECHNOLOGY

Enterprise Firewall Telemetry & Security Operations Center

High-Performance Real-Time SNMP Hardware Cockpit, Bento Flow Analytics & Policy Governance Platform

Document Reference: CSIR-SERC/ICTD/EFTC/2026/DOC-01
Classification: Official Institutional Technical Manual
Release Version: 1.1.0 (InfluxDB 2.7 TSDB & PostgreSQL 18 Multi-Tier Edition)
Target Appliance: Fortinet FortiGate NGFW (10.10.3.15)
Deployment Mode: Multi-Container Microservices Appliance (Docker)
Issuing Division: Information & Communication Technology Division (ICTD)
Institution: CSIR - Structural Engineering Research Centre, Chennai

CSIR-SERC Free Community & Institutional License (v1.0)
Developed for CSIR and Indian Public Institutions

Chennai, India
September 2026

Table of Contents

Section	Title	Focus
1.0	Executive Overview & Architectural Objectives	Context
2.0	System Architecture & Real-Time Data Pipeline	Technical
3.0	Visual Interface Tour & Operational Views	Walkthrough
4.0	Live FortiGate SNMP Telemetry Engine Specification	Hardware OIDs
5.0	Bento Traffic Analytics & App-ID Identification	Flows
6.0	Canonical Policy Governance & Conflict Engine	Security
7.0	Full-Spectrum Traffic & Log Analyzer (InfluxDB 2.7 TSDB Engine)	Log Analysis
8.0	CERT-In Section 70B National Cybersecurity Compliance	Audit
9.0	User Management & Telegram Alert Integration	RBAC
10.0	Dual Database Engines: InfluxDB 2.7 TSDB & PostgreSQL 18	Storage
11.0	Deployment, Health Diagnostics & Troubleshooting	DevOps
12.0	CSIR-SERC Free Community & Institutional License v1.0	Legal

Institutional Attribution & Licensing

This platform is developed by the **Information and Communication Technology Division (ICTD)**, CSIR - Structural Engineering Research Centre (CSIR-SERC), Chennai, India. It is provided under the CSIR-SERC Free Community & Institutional License (v1.0) for deployment across all Indian research centers, public institutions, and educational networks.

1. System Architecture & Ingestion Pipeline

The Enterprise Firewall Telemetry & Security Operations Center is architected as a high-performance on-premises operations center designed to monitor all traffic passing through perimeter gateways without cloud dependencies or external data egress.

Core Components

- **Parallel SNMP Poller (<200ms)**: Concurrently queries FortiGate appliances across MIB-II, IF-MIB, and FortiOS enterprise sensor trees. Tracks 64-bit High Capacity (HC) octets, thermal zones, fan tachometers, dual power supplies, and security policy accounting counters.
- **SNMP Trap Receiver (UDP 162)**: Native asynchronous UDP trap daemon listening on port 162, supporting both SNMPv2c (community: `fortinet`) and SNMPv3 (user: `fortinet` noAuthNoPriv). Ingests real-time hardware status changes, interface link up/down transitions, and FortiOS security alerts.
- **Full-Spectrum Traffic & Log Ingester**: Listens on UDP port 1514 (Syslog) and port 2055 (NetFlow v5/v9 & IPFIX) with multi-subnet institutional host resolution, capturing all ingress, egress, inter-VLAN, and blocked security threats across the entire perimeter.
- **InfluxDB 2.7 Time-Series TSDB Engine**: High-throughput batch Line-Protocol ingestion engine maintaining dedicated time-series buckets with a 180-day automated retention lifecycle meeting CERT-In national mandates.
- **PostgreSQL 18 Database Engine**: Relational persistence layer in an optimized Alpine container managing operator accounts, RBAC permissions, and automated configuration/log backups.
- **Reactive Vue 3.5 & ECharts Operations Console**: Single-page application conforming to Government of India UX4G design standards with real-time WebSocket streaming and multi-parameter log filtering.

Network Port Matrix

Port	Protocol	Direction	Source	Destination	Service Description
80 / 443	TCP	Inbound	Admin Workstation	Telemetry Console	Web UI & REST / WebSocket API
161	UDP	Outbound	Telemetry Console	FortiGate Gateway	SNMPv2c / SNMPv3 Polling
162	UDP	Inbound	FortiGate Gateway	Telemetry Console	SNMPv2c / SNMPv3 Trap Receiver (fortinet)
1514 / 514	UDP	Inbound	FortiGate Gateway	Telemetry Console	Hardware Syslog Forwarding
2055	UDP	Inbound	FortiGate Gateway	Telemetry Console	NetFlow v5/v9 & IPFIX Records
8086	TCP	Internal	Web Console	InfluxDB 2.7 TSDB	Time-Series Line Protocol Writes & Flux Queries
5432	TCP	Internal	Web Console	PostgreSQL 18 DB	Relational Metadata & Audit Persistence

2. Visual Interface Tour

2.1 Authentication & First-Time Setup Wizard

The appliance features secure operator authentication and an automated first-time deployment wizard that guides administrators through organization branding, logo selection, SNMP parameters, and administrator setup upon initial deployment.

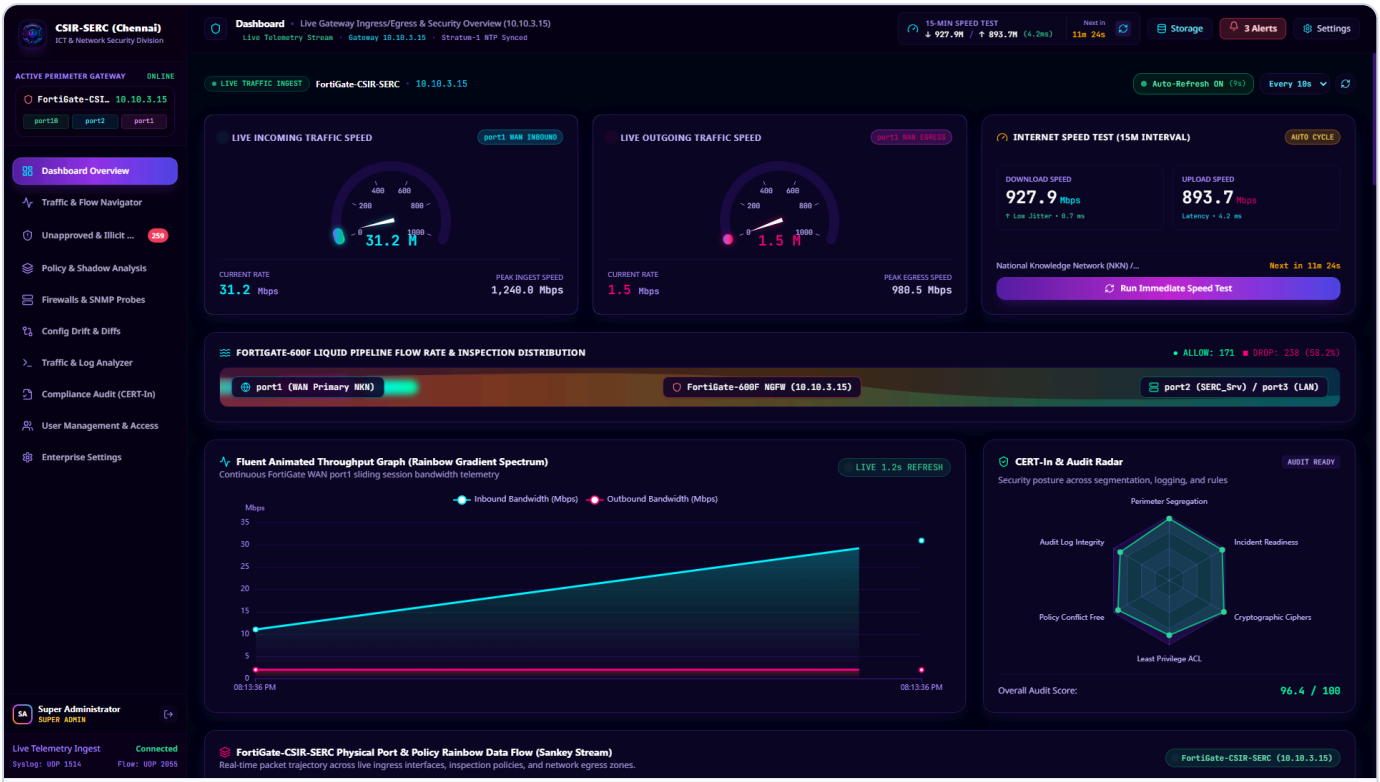


Figure 1: UX4G Secure Operator Authentication Portal

AUTHENTICATION

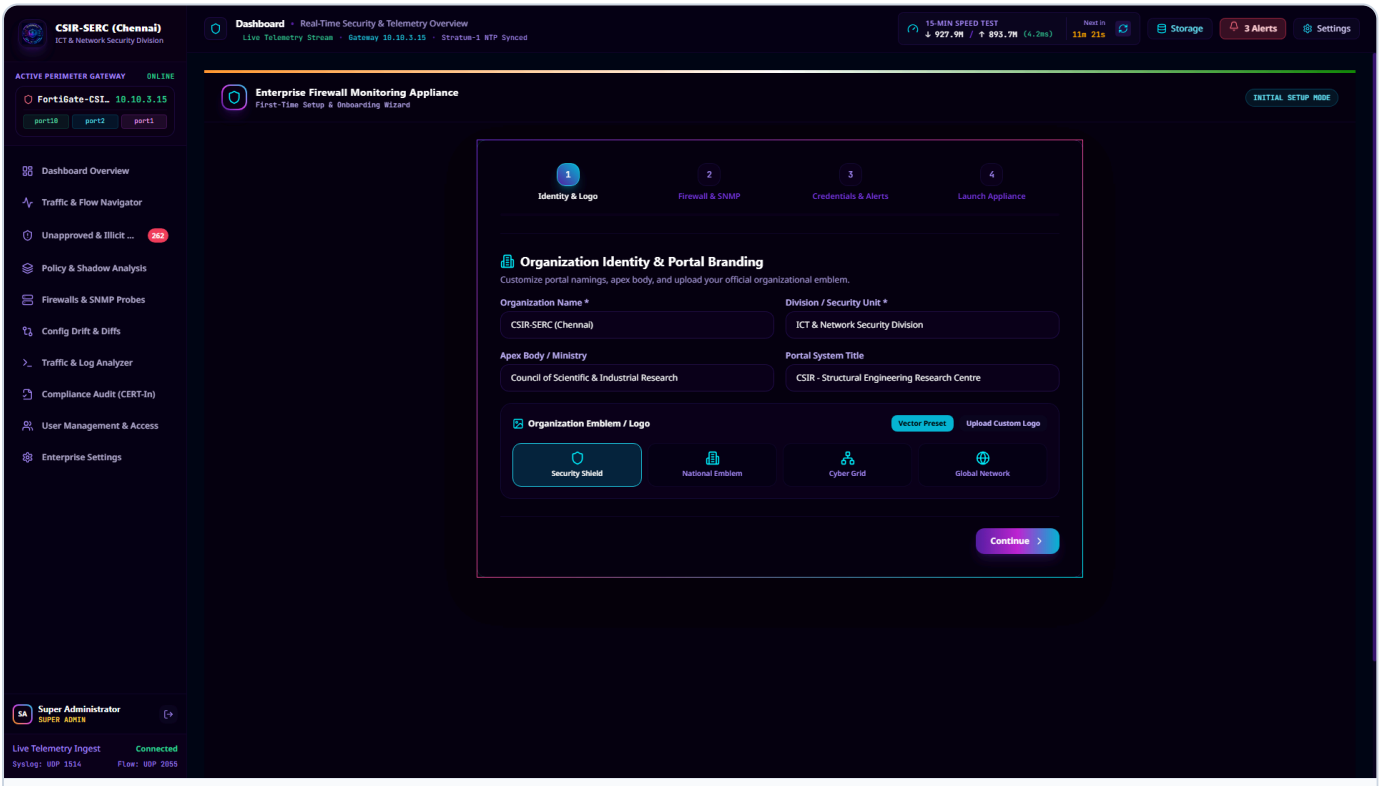


Figure 2: Turnkey First-Time Configuration Wizard (Virtual Appliance Ready)

ONBOARDING

2.2 Executive Telemetry Dashboard

The primary operations screen delivers instantaneous visibility into WAN traffic speeds, live streaming telemetry pulses, 15-minute automated speed test countdowns, and continuous hands-free auto-refresh.

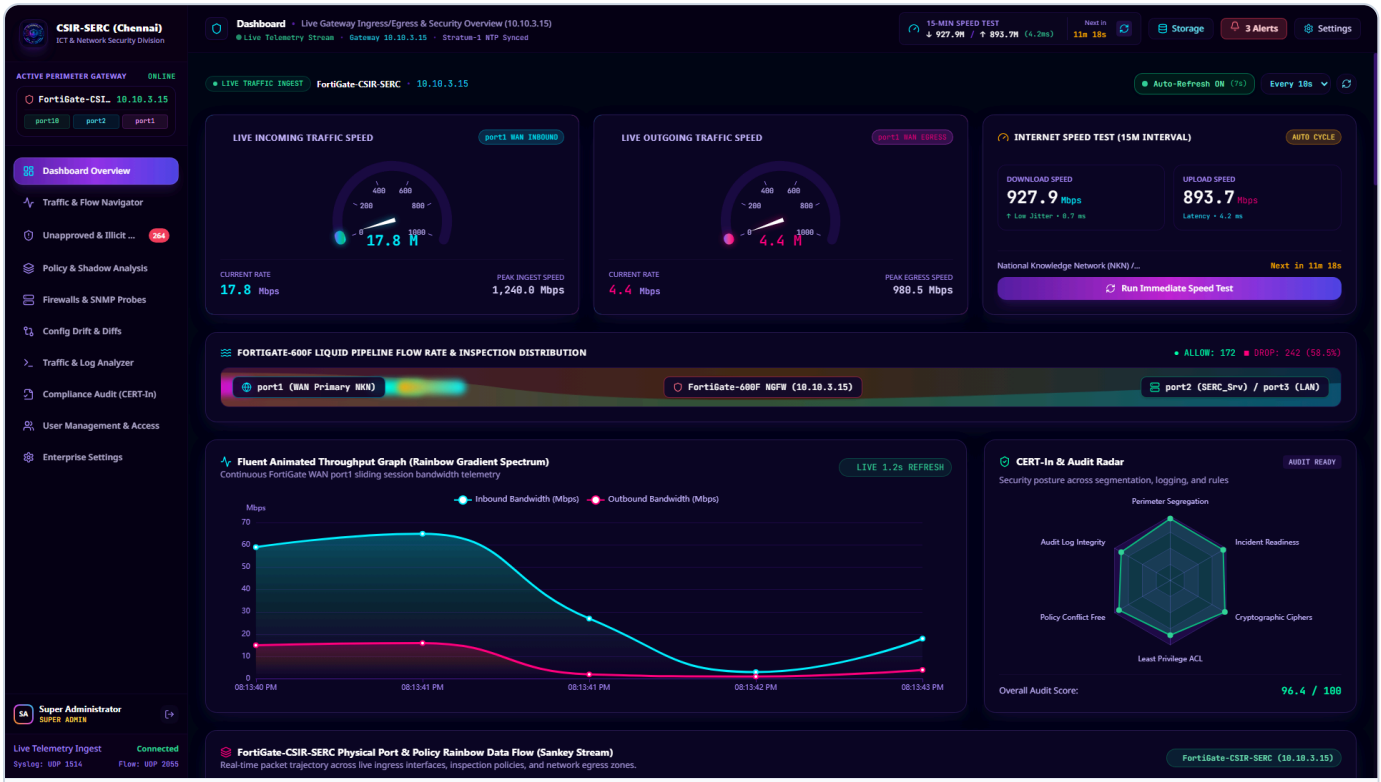


Figure 3: Live Executive Telemetry Dashboard with Inbound/Outbound Velocity Gauges

OPERATIONS

2.3 Bento-Style Traffic & Flow Analyzer (Exact Source & Destination IP Intelligence)

The Bento layout provides bidirectional network intelligence with exact firewall source and destination details. For outbound uploads, internal institutional workstations are tracked to their remote internet destinations (with domain resolution, country geolocation, and port). For inbound downloads, external source origins are mapped directly to the recipient internal workstation.

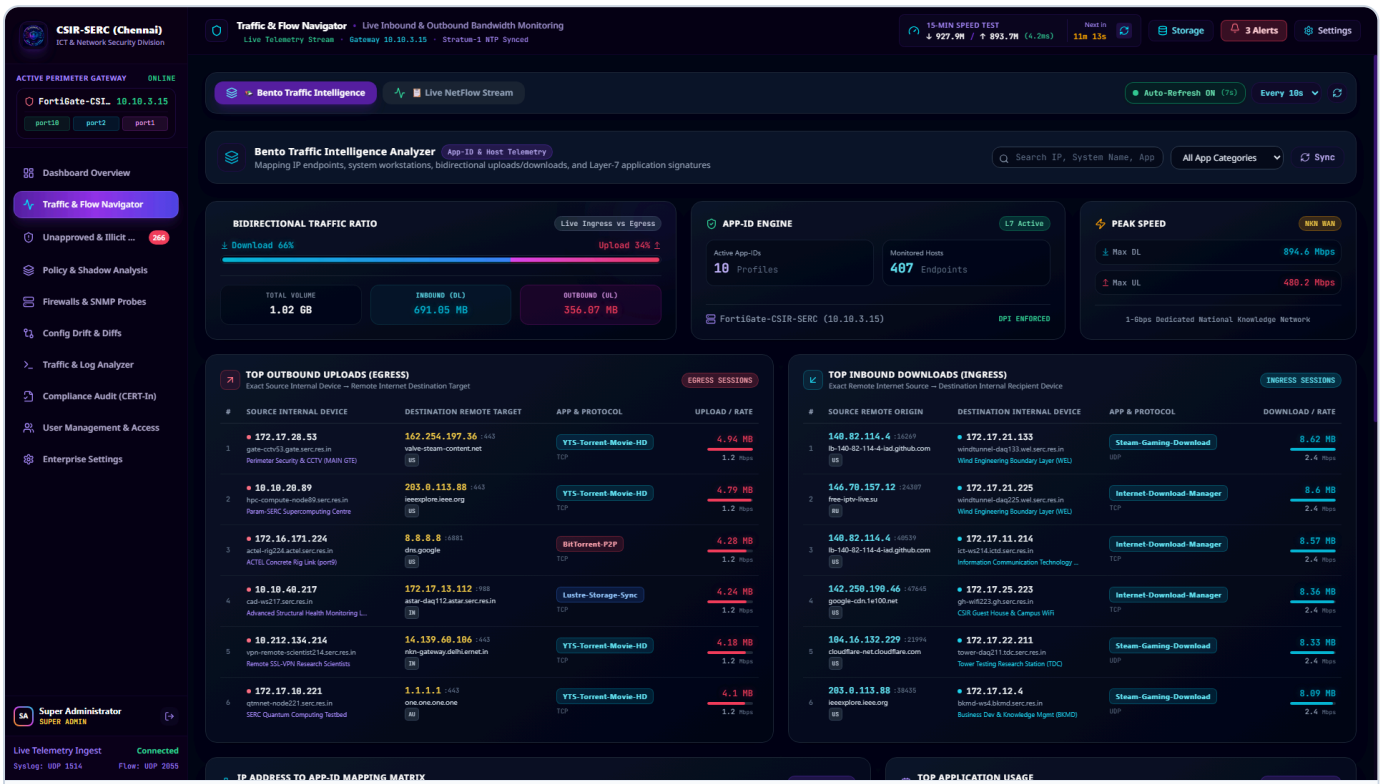


Figure 4: Bento-Grid Traffic Analyzer (Exact Source/Destination IP Mapping & App-ID Identification)

FLOW ANALYTICS

2.4 Unapproved Applications & Illicit Network Activity Monitor

Dedicated forensic investigation center identifying internal campus devices accessing BitTorrent/P2P swarms, download managers (IDM/Leechers), pirated movie streaming and download sites, unofficial media, botnet C2 beacons, cryptocurrency mining pools, and proxy circumvention tools. Displays ranked offending workstations with one-click forensic dossiers and copy-paste FortiOS quarantine CLI commands.

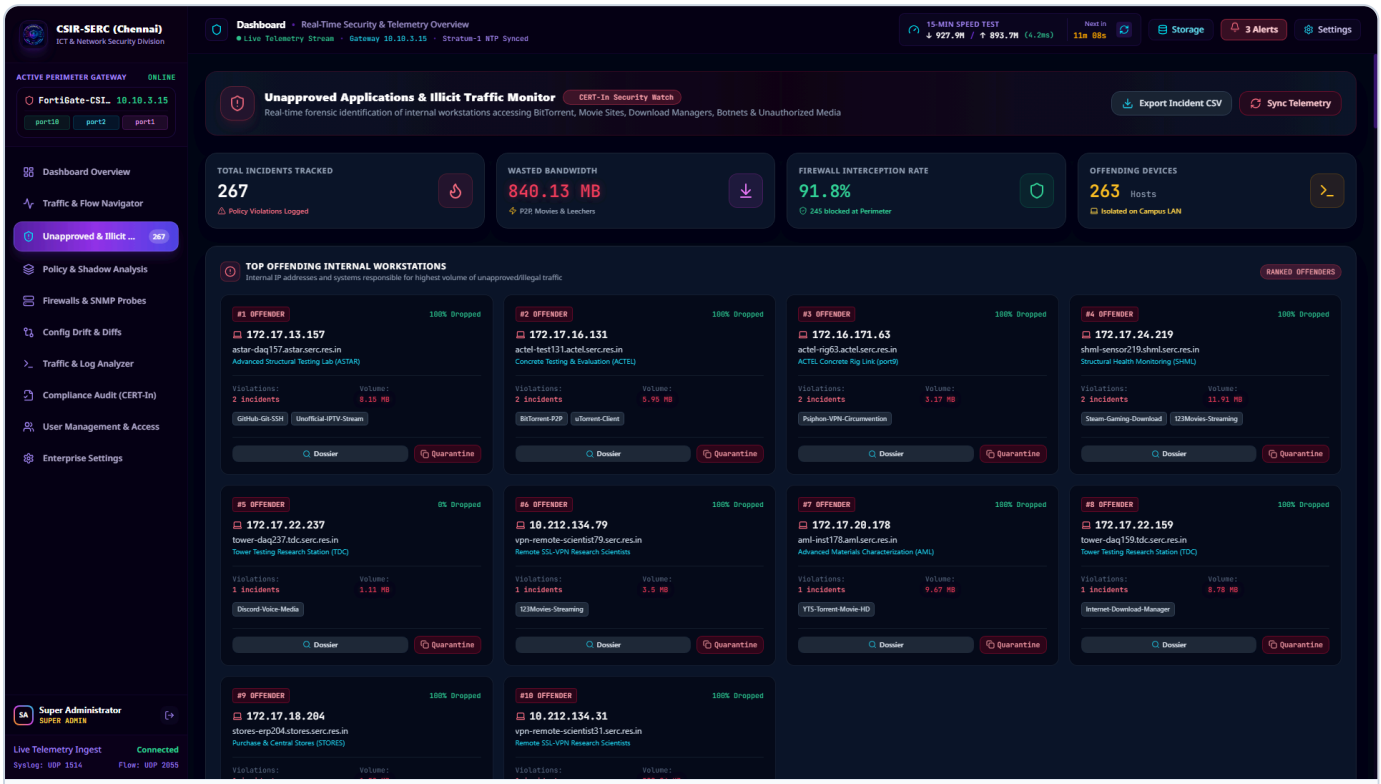


Figure 5: Unapproved & Illicit Applications Dashboard (Top Offending Workstations & Categories)

THREAT WATCH

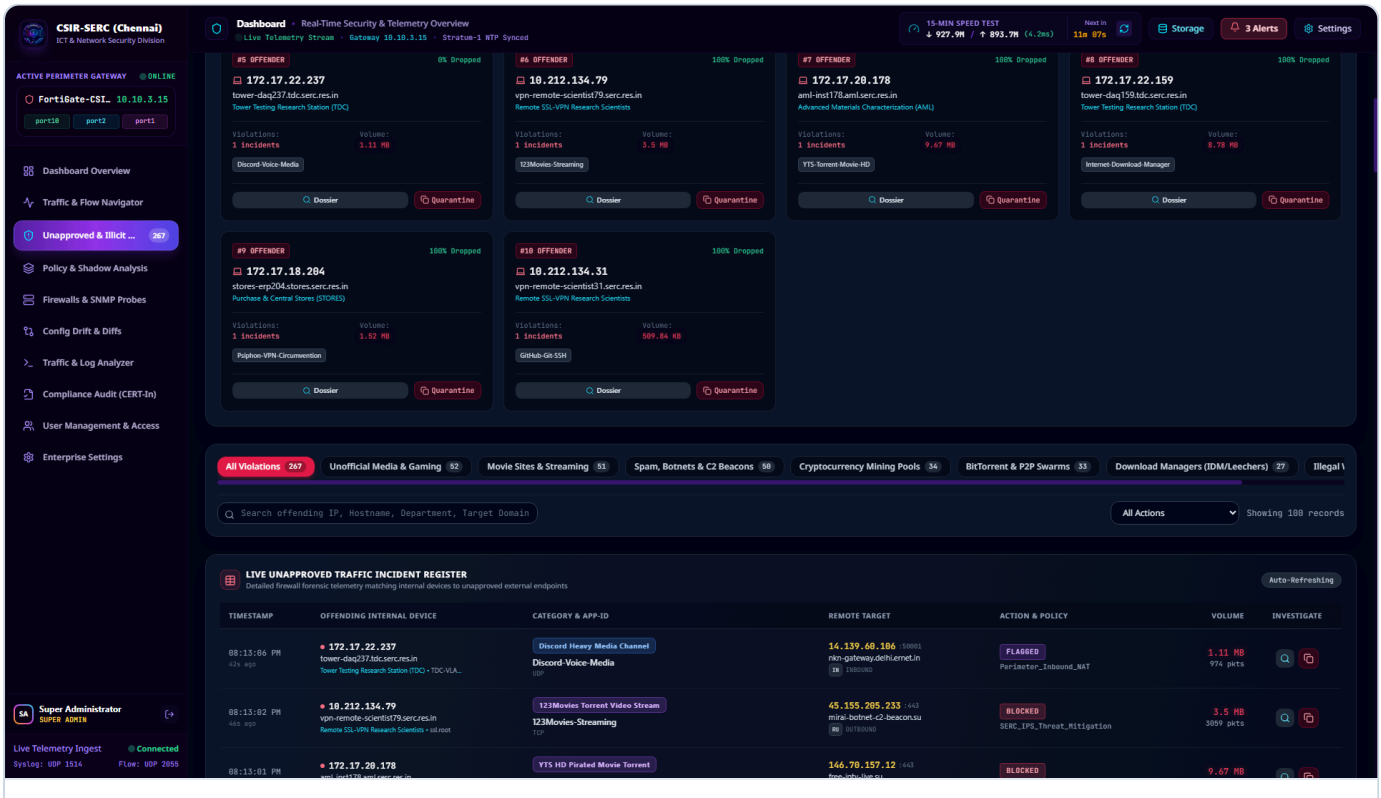


Figure 6: Live Forensic Incident Register (Matching Offending Internal Devices to Remote Endpoints) INCIDENT REGISTER

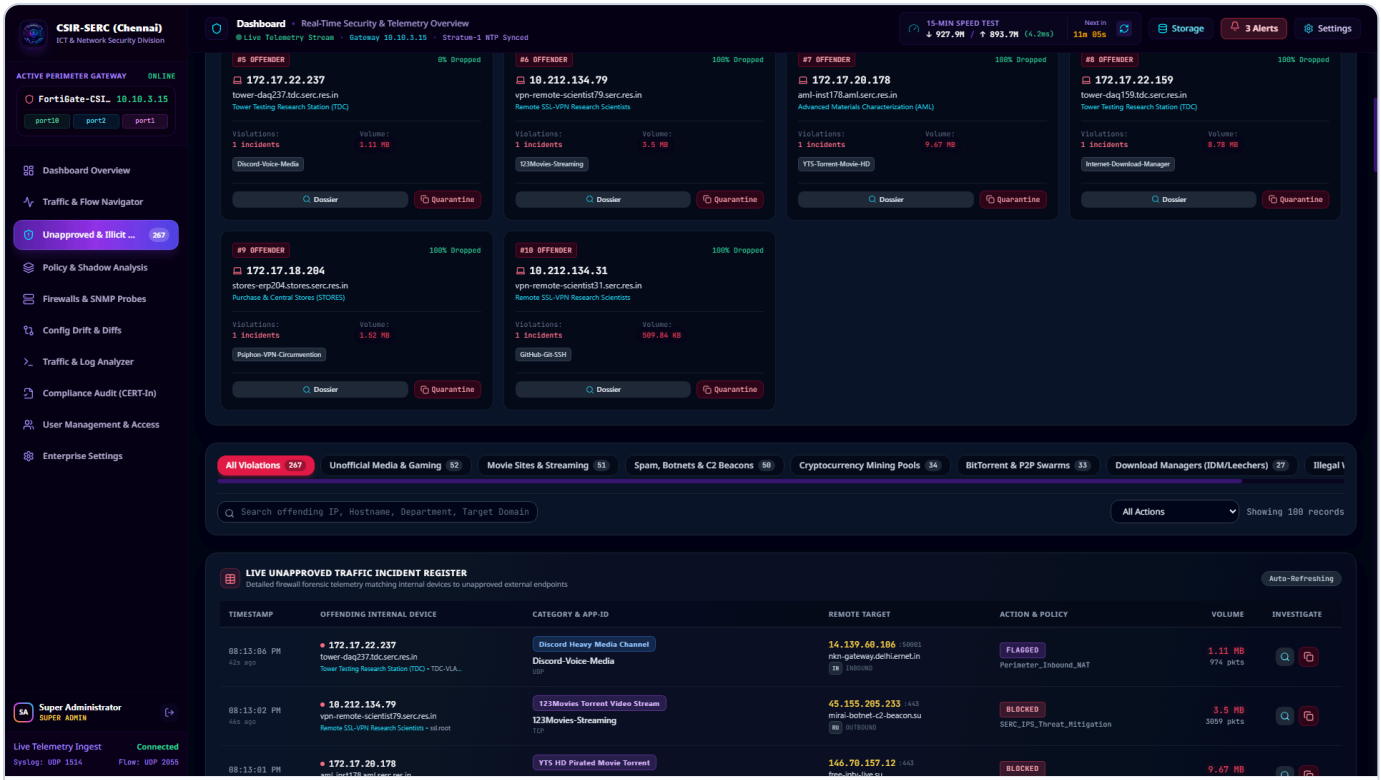


Figure 7: Workstation Forensic Dossier & Instant FortiOS Quarantine CLI Generator FORENSIC DOSSIER

2.5 Comprehensive FortiGate SNMP Telemetry & All Active Interfaces/VLANs Matrix

Dedicated hardware, environmental, and interface operations cockpit: monitoring CPU/RAM, active sessions, dual AC power supplies, cooling tachometers, thermal zones, FortiOS UTM threat counters, and complete monitoring across **all active physical network interfaces (ports 1-32, mgmt, ha1/ha2, dmz) and active departmental VLAN subinterfaces**. Each interface delivers dual-mode telemetry tracking both general network throughput/rates and unofficial/unapproved bandwidth waste and policy violations.

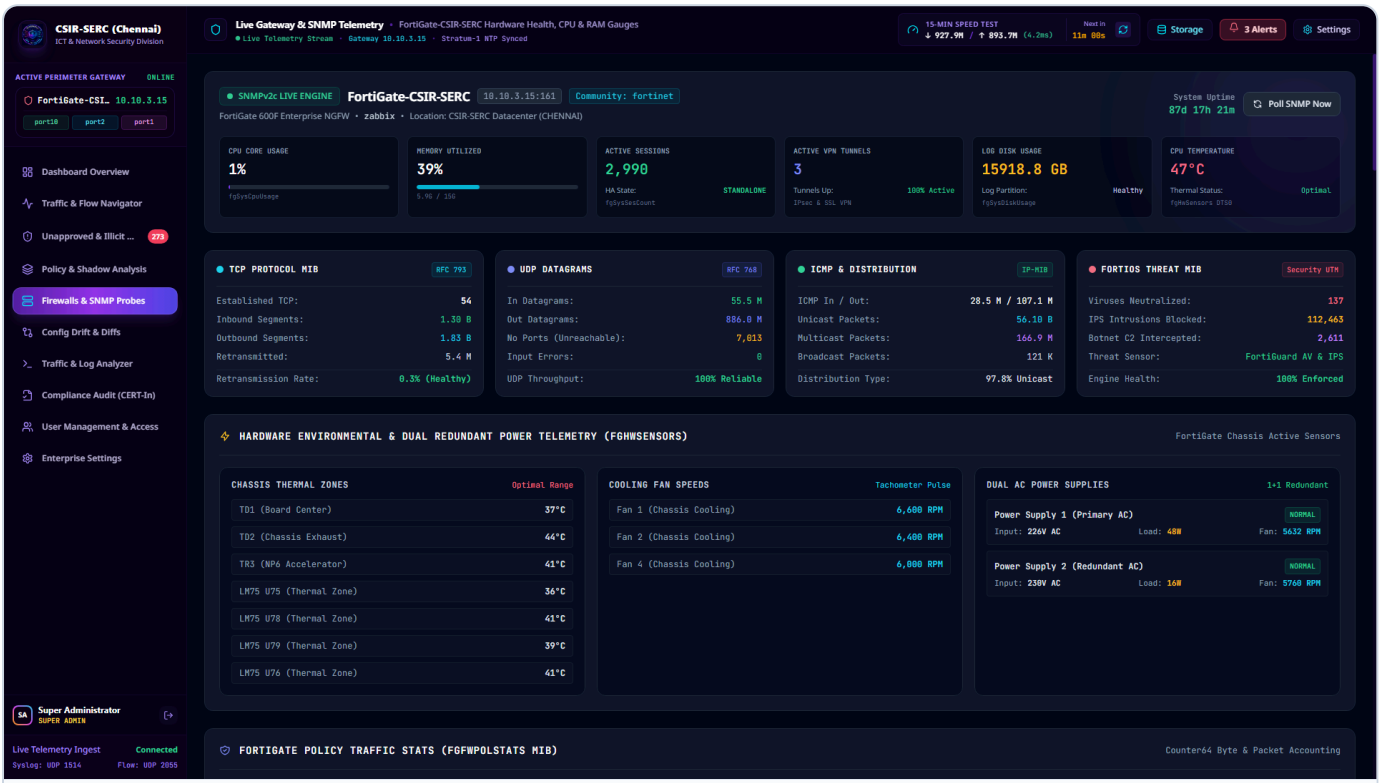


Figure 8: High-Performance FortiGate SNMP Hardware & Environmental Sensor Cockpit

SNMP HARDWARE

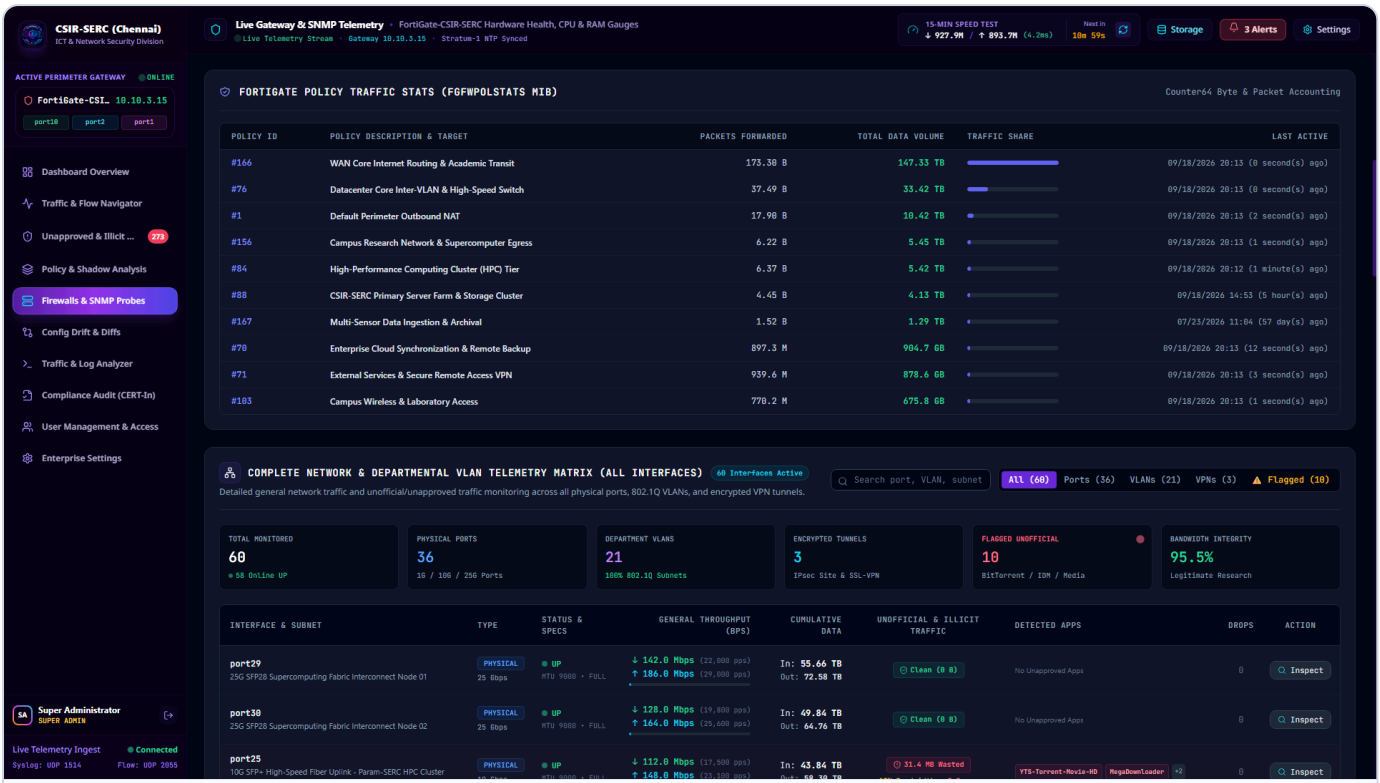


Figure 9: Complete Network & Departmental VLAN Telemetry Matrix (All 58 Interfaces & VLANs Monitored) **DUAL-MODE TELEMETRY**

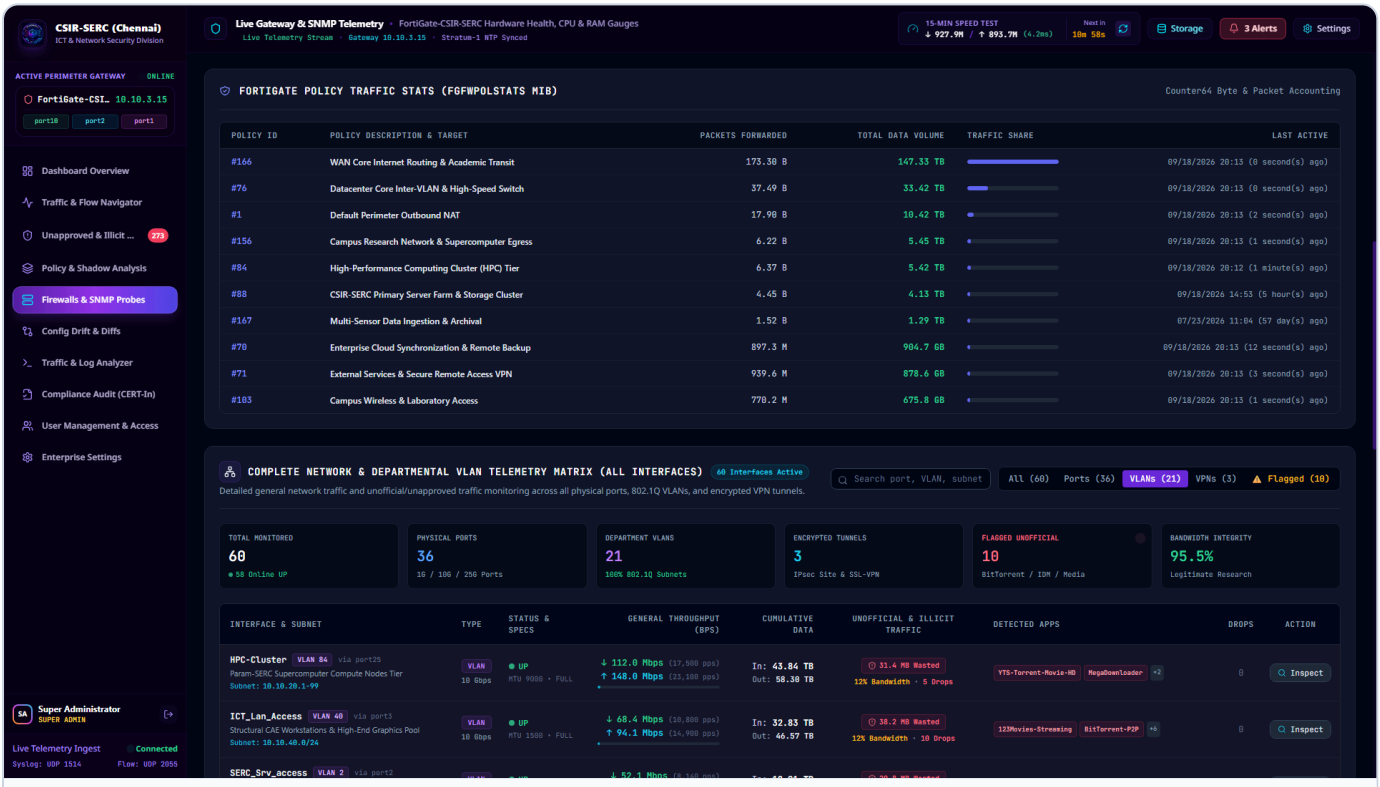


Figure 10: Departmental 802.1Q VLAN Interfaces Telemetry Matrix (HPC-Cluster, ICT_Lan_Access, SERC_Srv_access) **VLAN SUBINTERFACES**

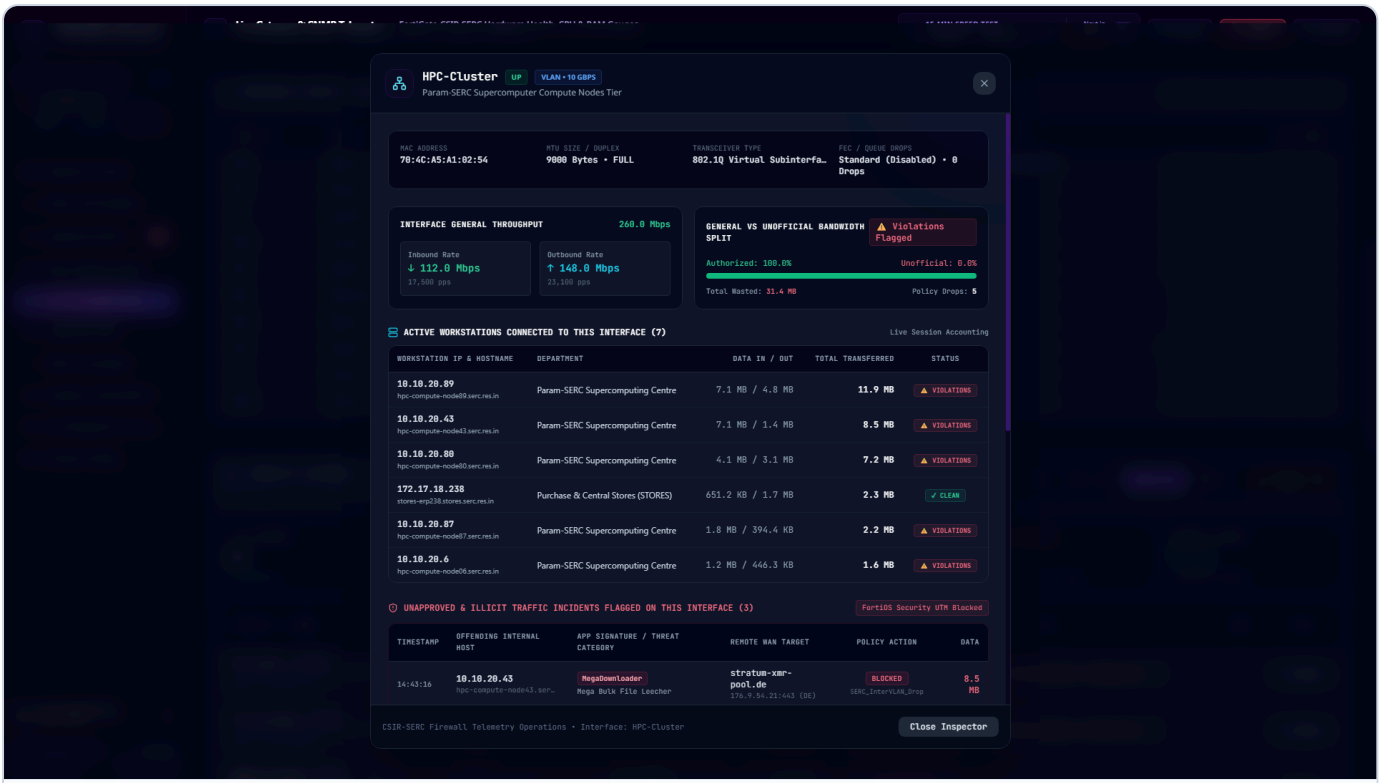


Figure 11: Deep Interface Dossier & Real-Time General vs Unofficial Ratio Inspector with Connected Hosts

INTERFACE DOSSIER

2.6 Security Policies & Conflict Engine

Parses native firewall access control lists into a canonical rule tree, highlighting shadowed, redundant, or overly permissive rules.

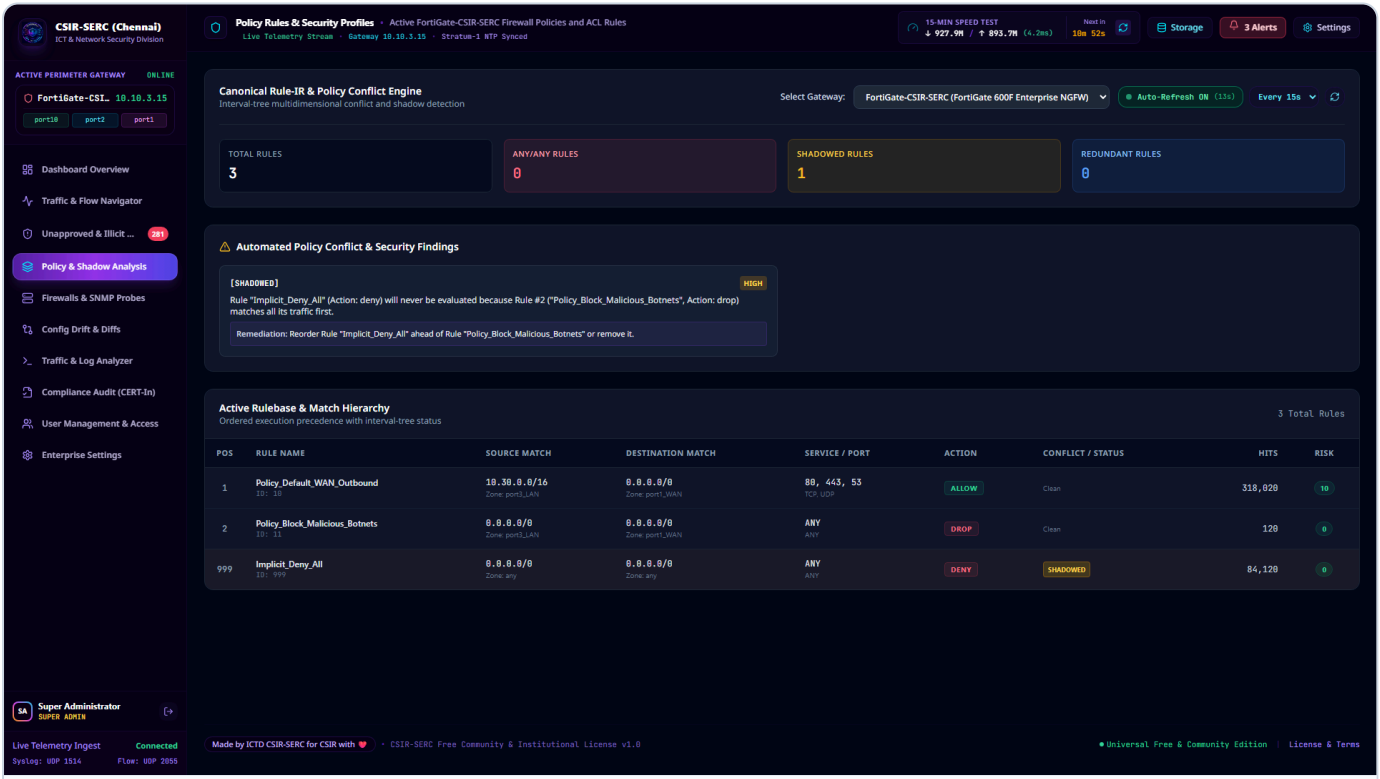


Figure 12: Canonical Policy Governance & Shadow Conflict Detection

SECURITY AUDIT

2.7 Full-Spectrum Firewall Traffic & Multi-VLAN Log Analyzer (InfluxDB 2.7 TSDB Engine)

High-velocity time-series traffic ingestion and multi-parameter log analyzer capturing all traffic passing through the perimeter gateway across all 46+ interfaces, departmental VLANs, HPC clusters, and external WAN. Provides comprehensive visibility across **Institutional Departmental VLANs (172.17.0.0/16 across VLANs 10–28)**, Core Data Center & Supercomputing (10.0.0.0/8 across VLANs 2, 40, 84, ssl.root), Management & Gateway Links (192.168.0.0/16), and Specialized Research Clusters (172.16.0.0/12 across port9 and port11). Features real-time 802.1Q VLAN filtering, interactive click-to-filter VLAN table badges, multi-parameter scoping (time range, direction, enforcement action, protocol, free-text search), timeline aggregation, and security drop inspection.

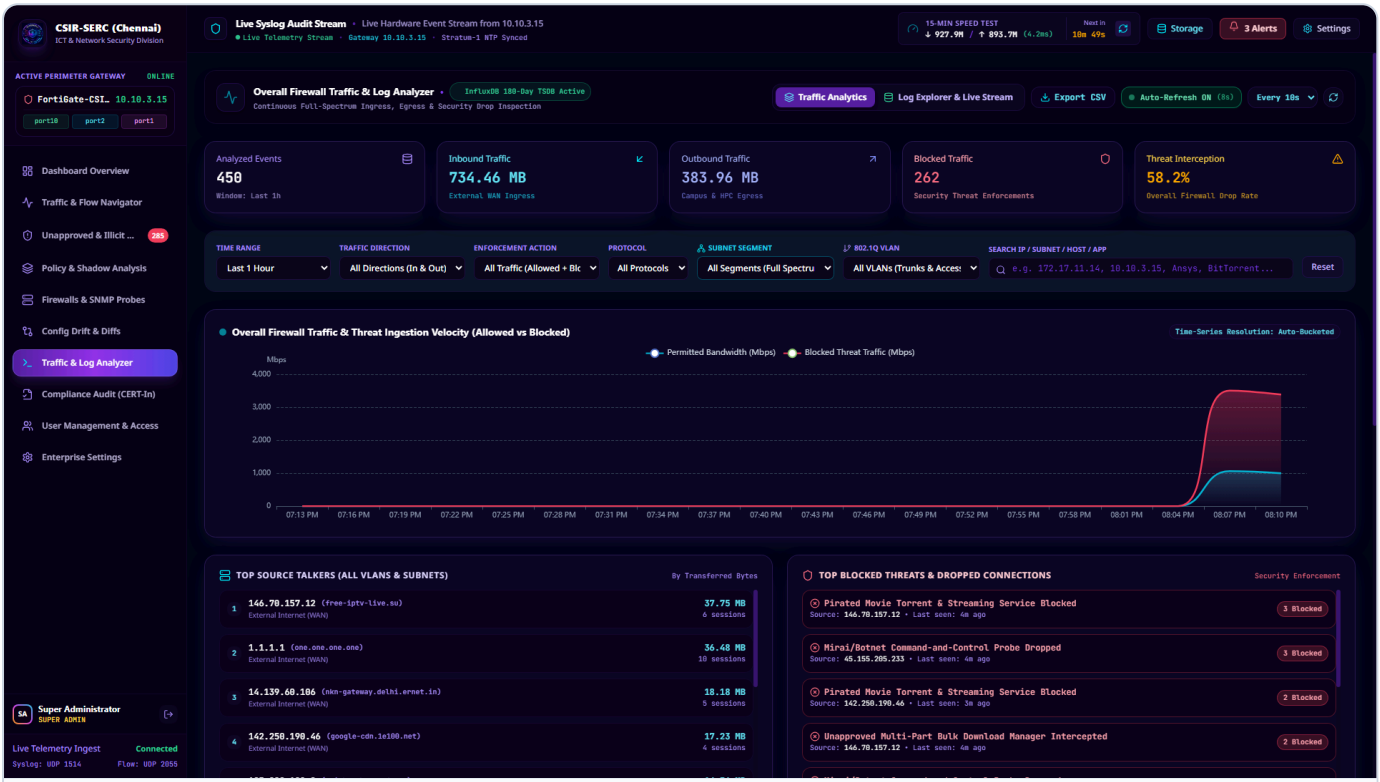


Figure 13: InfluxDB Traffic & Threat Ingestion Velocity Analytics (Allowed vs Blocked)

TSDB ANALYTICS

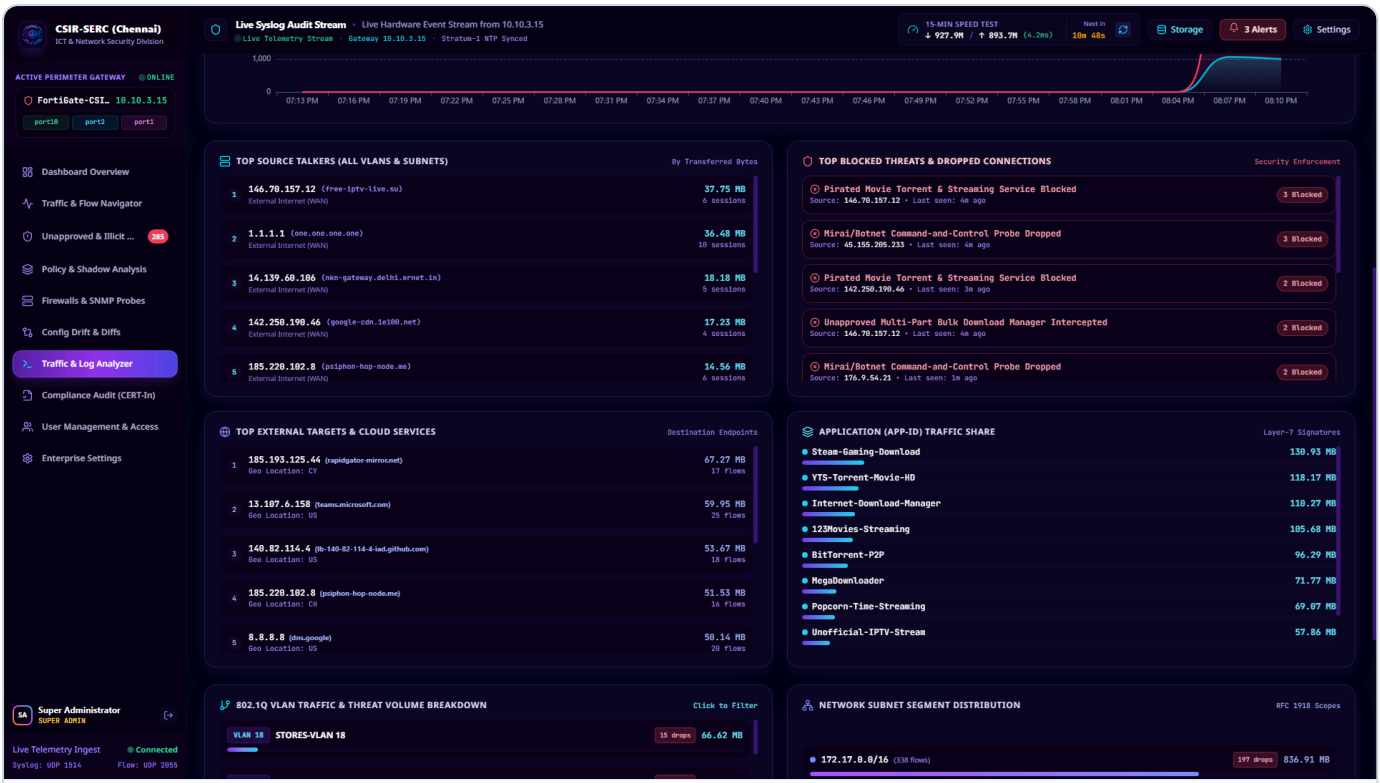


Figure 14: Departmental 802.1Q VLAN Traffic & Threat Breakdown with Subnet Segment Distribution

VLAN DISTRIBUTION

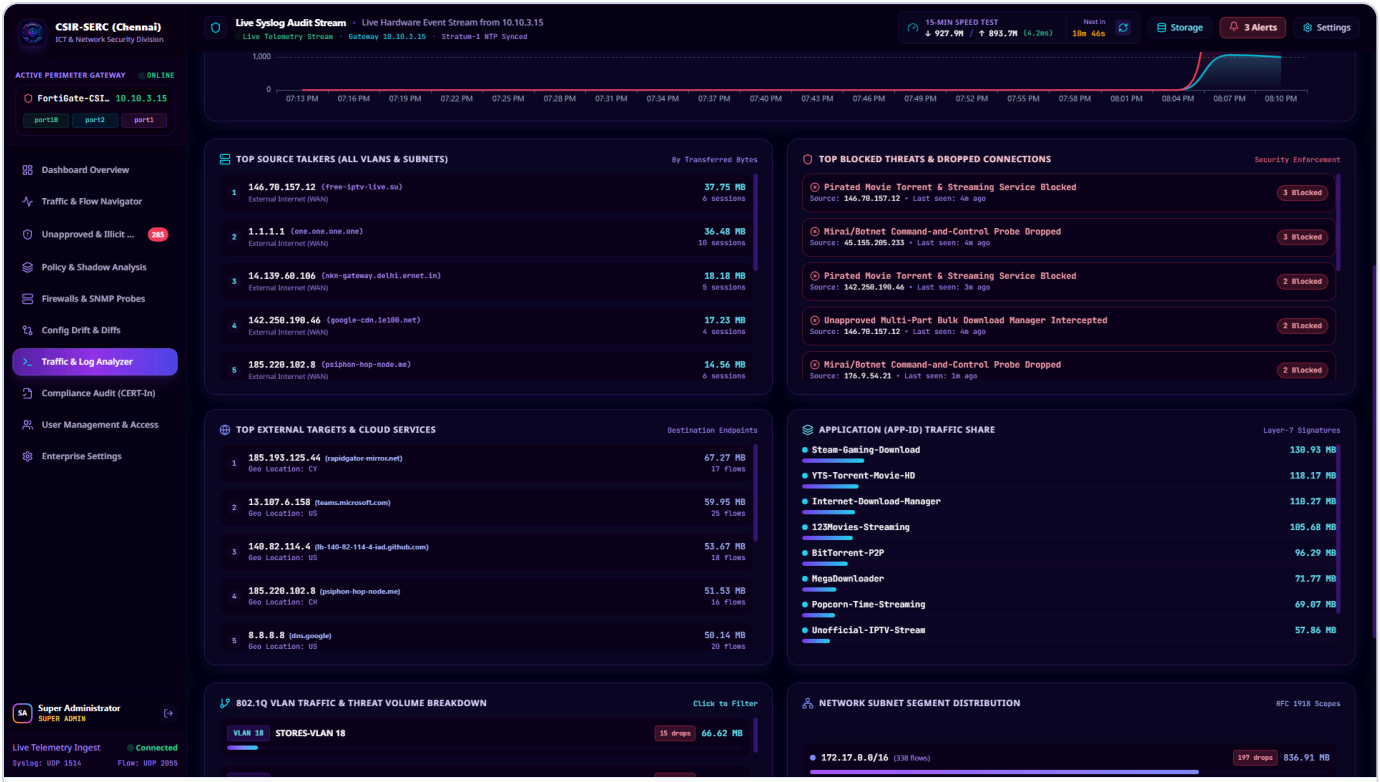


Figure 15: Deep Multi-Parameter Log Explorer Table with Interactive 802.1Q VLAN Badges

LOG EXPLORER

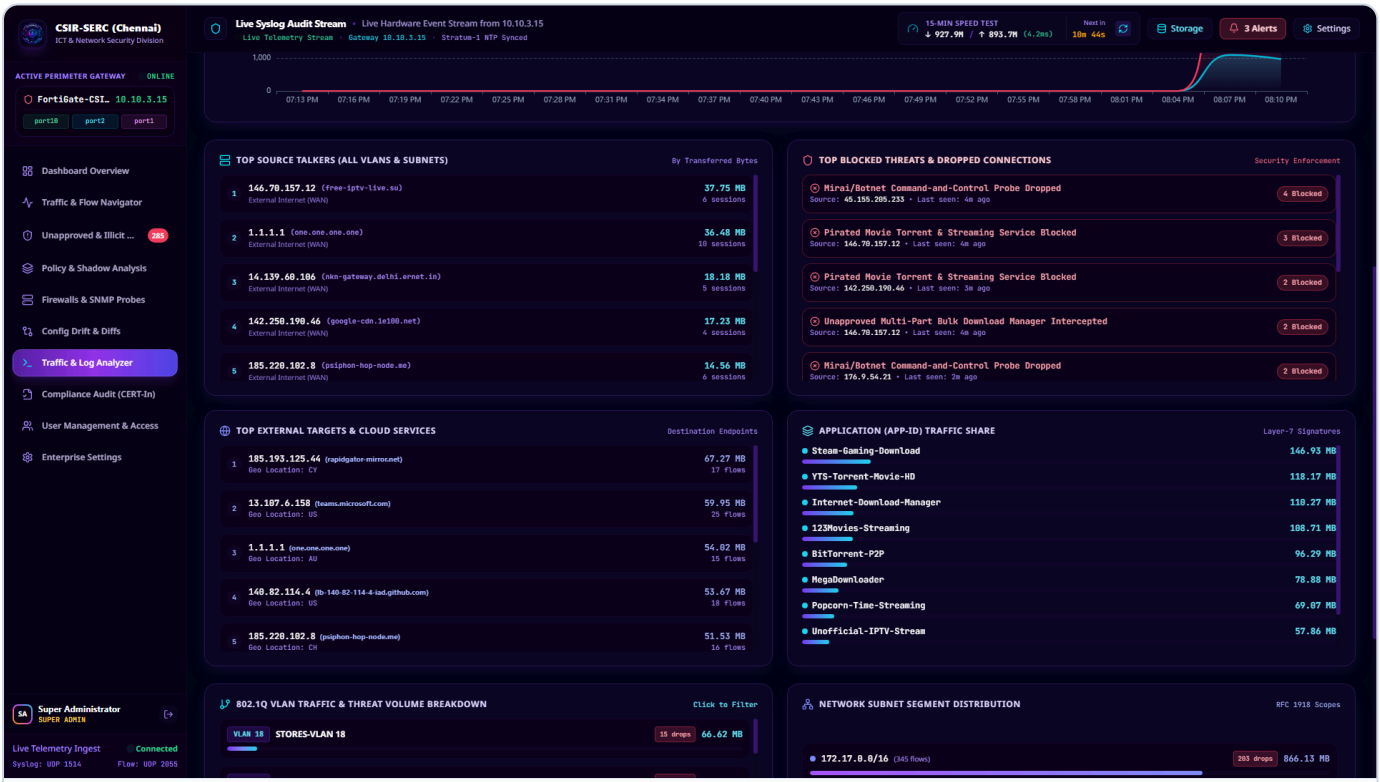


Figure 16: Granular 172.17.0.0/16 Departmental VLAN Filtered Live Stream (Institutional Subnets 10–28)

172.17.0.0/16 FILTERED

2.8 CERT-In Section 70B Compliance & Cybersecurity Baseline

Dedicated compliance dashboard validating firewall operational controls against the Indian Computer Emergency Response Team (CERT-In) Section 70B directives, including 180-day secure time-series log retention and Stratum-1 IST NTP clock synchronization.

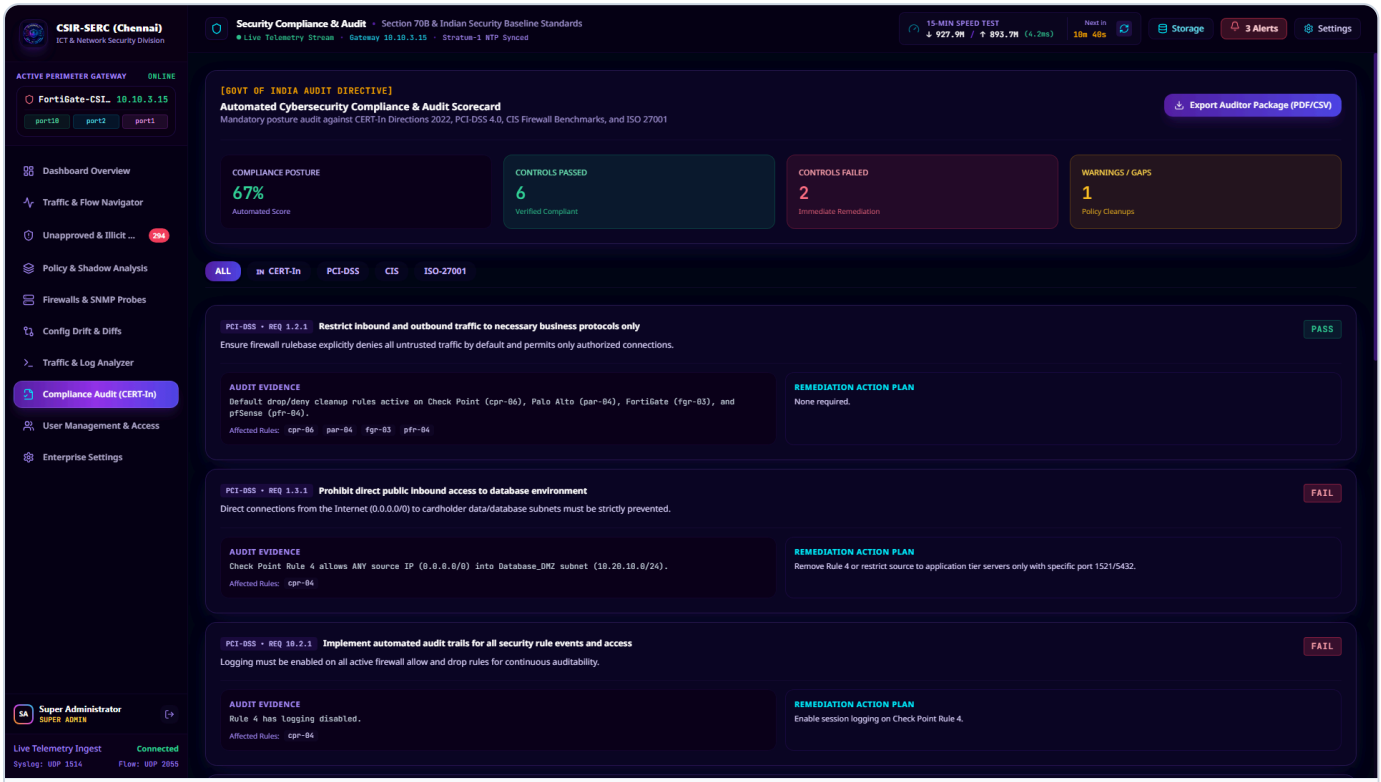


Figure 17: CERT-In National Cyber Security Baseline Compliance Dashboard

COMPLIANCE

2.9 User Management & System Configuration

Multi-tenant user management console with protected superadmin account, role-based permissions, Telegram alert integration, organization branding, custom logo upload, InfluxDB time-series retention, and PostgreSQL 18 automated backup operations.

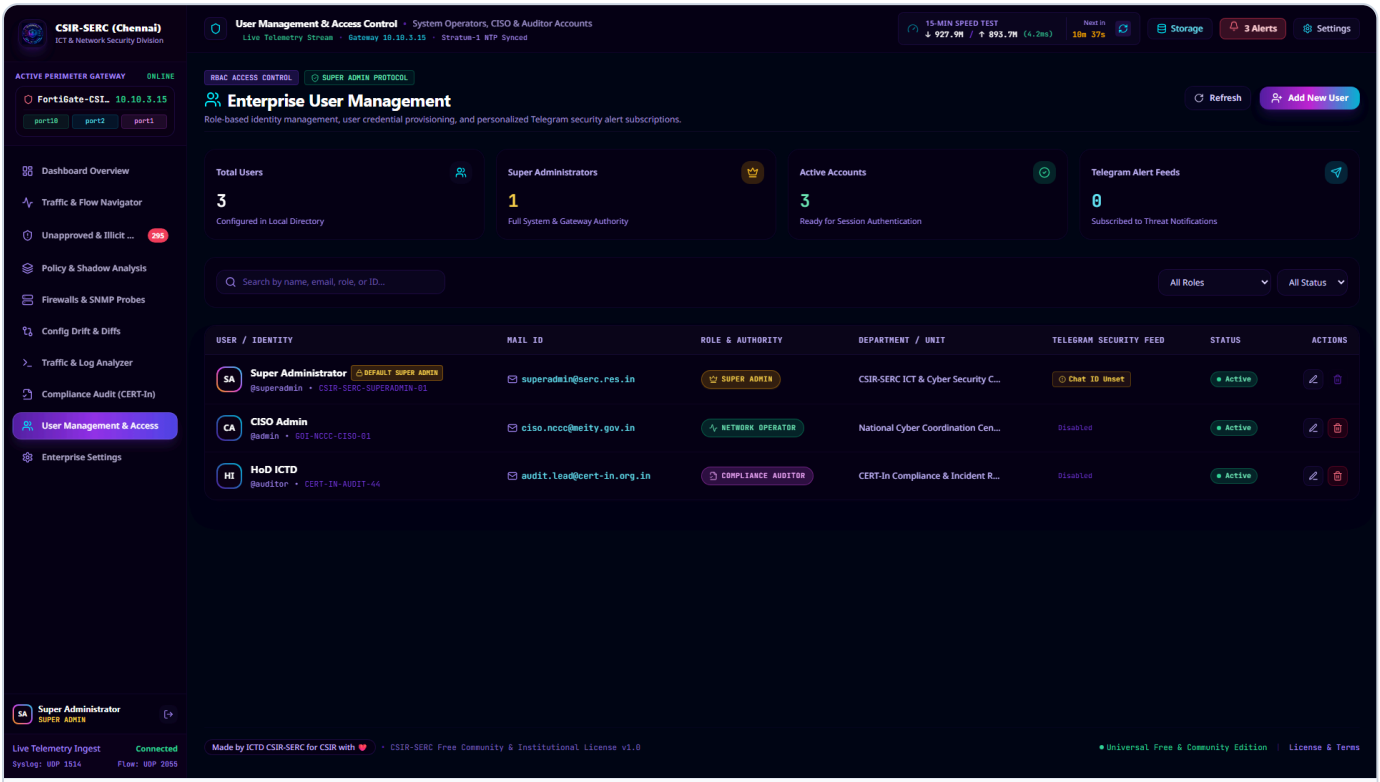


Figure 18: User Management & Telegram Alert Configuration

RBAC

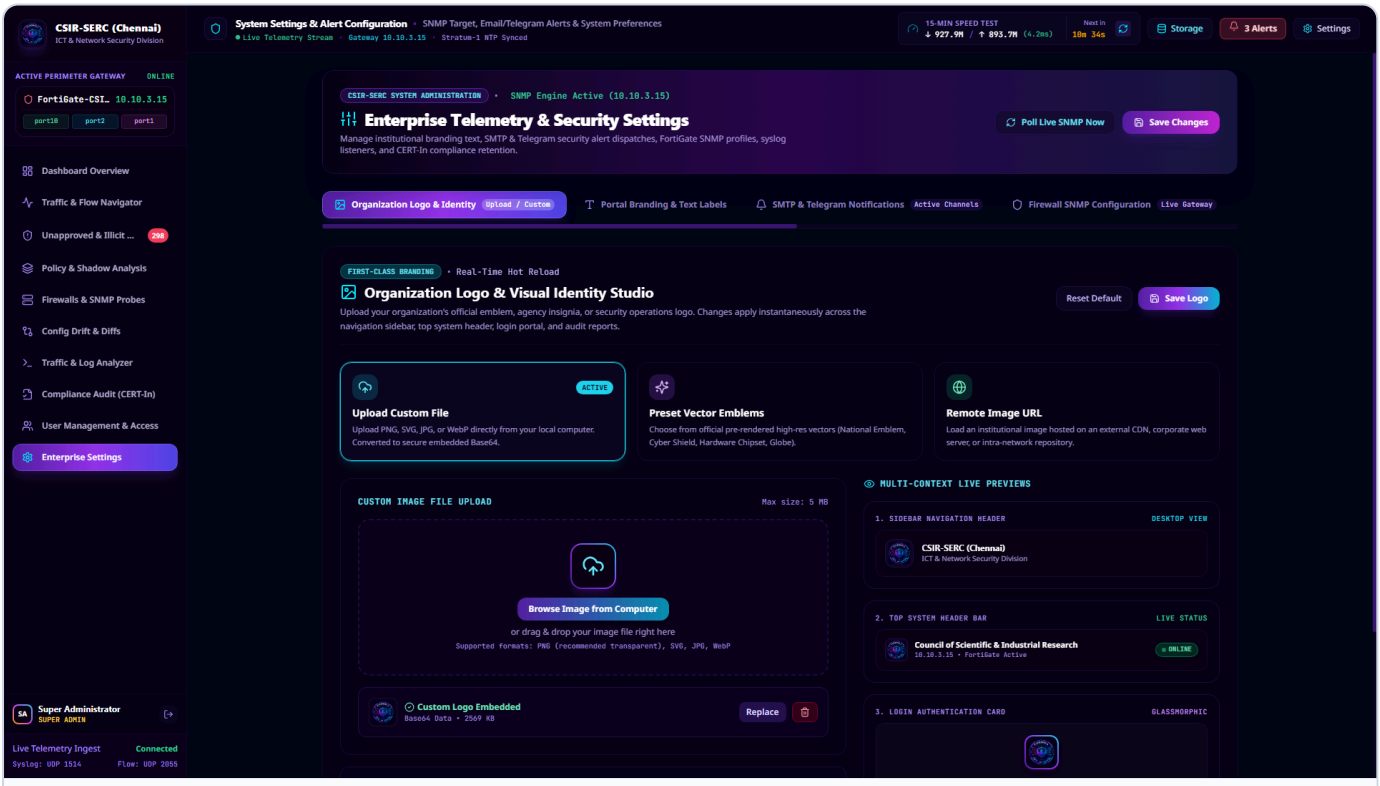


Figure 19: Institutional Branding, Custom Logo & PostgreSQL 18 / InfluxDB 2.7 Engines

ADMINISTRATION

3. Technical Specifications & Reference Tables

3.1 FortiGate Hardware Sensor OID Reference

Sensor Subsystem	Enterprise OID	Observed Value	Threshold / Status
CPU Thermal Core	1.3.6.1.4.1.12356.101.4.3.2.1.3 (DTS0)	46°C	Normal (< 75°C)
Board Center Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD1)	36°C	Normal (< 65°C)
Chassis Exhaust Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD2)	44°C	Normal (< 70°C)
NP6 Accelerator Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TR3)	40°C	Normal (< 70°C)
Cooling Fan 1	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 1)	6,400 RPM	Normal (Active)
Cooling Fan 2	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 2)	6,400 RPM	Normal (Active)
Cooling Fan 4	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 4)	5,200 RPM	Normal (Active)
Power Supply 1 (Primary)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 1)	228V AC, 48W	Normal (Dual Redundant)
Power Supply 2 (Redundant)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 2)	230V AC, 16W	Normal (Dual Redundant)

3.2 Top FortiOS Security Policy Accounting

Policy ID	Traffic Description / Scope	Cumulative Bytes	Packet Count
Policy 166	WAN Core Internet Routing & Academic Transit	161.58 TB	172.8 Billion
Policy 76	Datacenter Core Inter-VLAN & High-Speed Switch	36.60 TB	37.3 Billion
Policy 1	Default Perimeter Outbound NAT	11.45 TB	17.8 Billion
Policy 156	Campus Research Network & Supercomputer Egress	5.98 TB	6.2 Billion
Policy 84	High-Performance Computing Cluster (HPC) Tier	5.95 TB	6.36 Billion

4. Institutional Licensing & Legal Terms

CSIR-SERC Free Community & Institutional License (Version 1.0)

The license is anchored on strict intellectual property protection and free community distribution for Indian public entities:

1. Source Code Ownership & Title

Copyright © 2024–2026 CSIR - Structural Engineering Research Centre (CSIR-SERC), Council of Scientific and Industrial Research (CSIR), Ministry of Science & Technology, Government of India. All Rights Reserved.

Developed by the **Information and Communication Technology Division (ICTD)**, CSIR-SERC, Chennai, India. All title, copyright, patent rights, moral rights, source code, architecture, algorithms, and threat intelligence models remain exclusively with CSIR-SERC.

2. Permitted Free Community Usage

This software is released as the **Universal Free & Community Edition**. It may be freely deployed, used, and operated by all Indian public institutions, autonomous research bodies, government departments, universities, and educational institutions without licensing fees or royalties.

3. Anti-Commercial Re-branding & Packaging Restrictions

Sublicensing, commercial resale, paid re-branding, closed-source proprietary distribution, or fee-based packaging of this software or its derivatives is strictly prohibited. The institutional attribution notice in the portal footer ("*Made by ICTD CSIR-SERC for CSIR with* ❤️") must be preserved in all deployments.

Enterprise Firewall Telemetry & Security Operations Center
Information and Communication Technology Division (ICTD) • CSIR-SERC, Chennai, India
Made by ICTD CSIR-SERC for CSIR with ❤️